DataSentinel – Intelligent Data Quality & Analysis System

Project Introduction

In real-world systems, data is rarely clean. Organizations constantly deal with incomplete, inconsistent, and unreliable datasets, which directly affect decision-making and analytics. Writing scripts to clean and analyze such data requires technical expertise and time, making it inefficient for non-technical users.

To solve this problem, **DataSentinel** was designed as an intelligent, browser-based platform that automates the entire data preparation pipeline. Instead of manually writing code, users can upload raw datasets and seamlessly move through a structured pipeline: **profiling, cleaning, and analysis**.

The system not only detects data quality issues but also intelligently suggests and applies corrections using multiple strategies with retry mechanisms. Every transformation is tracked through versioning and data lineage, ensuring transparency and traceability. Finally, the platform generates meaningful insights and interactive visualizations to support data-driven decisions.

Core Features & Highlights

1. Automated Data Profiling

- Detects missing values, duplicates, outliers (IQR method), and invalid entries
- Identifies data type mismatches and inconsistencies
- Generates detailed statistical summaries

2. Intelligent Data Cleaning Engine

- Supports **12 cleaning strategies** (mean, median, mode, forward fill, etc.)
- Implements **retry logic (up to 10 attempts)** for robust cleaning
- Verifies success using reduction thresholds (90% missing, 80% outliers)

3. Data Versioning & Lineage

- Maintains **snapshot history (v1, v2, v3…)** after each transformation
- Tracks every operation with execution status and time
- Enables rollback and auditability

4. Exploratory Data Analysis (EDA)

- Generates summary statistics and KPIs
- Builds **correlation matrices**
- Supports interactive charts using Plotly (bar, line, scatter, etc.)

5. User-Friendly Interface

- No coding required
- Dark-themed modern UI with real-time feedback
- Dataset management and issue filtering system

Technical Architecture

- **Backend:** Django 4.2
- **Data Processing:** Pandas, NumPy, SciPy
- **Frontend:** HTML, CSS, JavaScript, Plotly.js
- **Database:** SQLite

Key Technical Strengths

- Modular design using **DataProfiler, CleaningEngine, and EDAEngine**
- Robust **retry + verification mechanism** for cleaning
- Efficient handling of structured datasets (CSV/XLSX)
- Secure system with local processing (no external API dependency)
- Scalable architecture with potential for async processing (Celery)

Project Links

- **GitHub Repository:**
  https://github.com/codexcherry/DataSentinel
- **Live Deployment (Domain):**
  https://nithinvs.site
- **AWS Deployment (Direct Access):**
  http://16.171.237.105:8000

Conclusion

DataSentinel transforms raw, unreliable data into structured, clean, and analyzable information through an intelligent and automated pipeline. By combining data engineering principles with a user-friendly interface, the system bridges the gap between technical complexity and practical usability, making data analysis accessible, efficient, and reliable.